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MILITARY SPECIFICATION

LASER INFRARED OBSERVATION SET, AN/GVS-5

This specification is approved for use by the USA Electronic Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of laser rangefinder designated as Laser Infrared Observation Set AN/GVS-5 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal (except as noted in paragraph 6 of this specification) form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-G-174	- Glass, Optical
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	- Marking of Electronic Items
MIL-F-14072	- Finishes for Ground Electronic Equipment
MIL-P-15024/6	- Plastic Molding Material (Polystyrene Foam Expanded Bead)
MIL-P-19644	- Plastic Molding Material (Polystyrene Foam Expanded Bead)
MIL-P-26514	- Polyurethane Foam, Rigid or Plastic, for Packaging
MIL-E-55585	- Electronic Equipment and Parts, Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, USA Electronics Research and Development Command, ATTN: DRDEL-E, 2800 Powder Mill Road, Adelphi, MD 20783 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5860

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	- Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurements of
MIL-STD-726	- Packaging Requirement Code
MIL-STD-781	- Reliability Tests Exponential Distribution
MIL-STD-810	- Environmental Test Methods
MIL-STD-1275	- Electrical Circuit, 28 Volt DC Transient Characteristics for Military Vehicles
MIL-STD-1474	- Noise Limits for Army Materiel

DRAWINGS

US ARMY ELECTRONICS COMMAND

DL SM-B-852100	- Laser Infrared Observation Set

FEDERAL PUBLICATIONS

Code of Federal Regulations, Title 10 Schedule A., Energy

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

United States of America Standards Institute

S1.4	- General Purpose Sound Level Meters
S1.11	- Octave, Half-octave and Third Octave Band Filter Sets

(Application for copies shall be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018).

3. REQUIREMENTS

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3.1 Description. The Laser Infrared Observation Set, AN/GVS-5 herein referred to as LR Set, is a laser distance measuring set in a watertight case. Laser Infrared Observation Device MX-9838/GVS-5, herein referred to as LR, the principal unit, is a lightweight, individually-operated, hand-held laser rangefinder capable of determining the range to targets between the specified minimum and maximum range. In addition the set contains Special Purpose Electrical Cable Assembly CX-13021/GVS-5 for operation of the laser rangefinder from external power sources. An adapter bracket, Mounting MT-4832/GVS-5 for operation in conjunction with a night vision device is also included.

3.2 Construction. The LR Set shall be constructed in accordance with the requirements of this specification and DL-SM-B-852100.

3.2.1 Weight. The maximum total weight of the Laser Infrared Observation Set AN/GVS-5 shall not exceed 37.6 pounds (17.1 Kg). Weights of the indivudual units shall be as indicated below:

Unit	Weight
Laser Infrared Observation Device MX-9838/GVS-5 (Including internal battery BB-516/U)	5.0 lbs (2.3 Kg) maximum
Cable Assembly, Special Purpose, Electrical CX-13021/GVS-5	2.2 lbs (1.0 Kg) maximum
Mount Adapter MT-4832/GVS-5	2.0 lbs (0.9 Kg) maximum
Case, Field, Laser Infrared Observation Device CY-7536/GVS-5	4.4 lbs (2.0 Kg) maximum
Case, Transit, Laser Infrared Observation Set CY-7535/GVS-5	24.0 lbs (10 Kg) maximum

3.3 First article. When specified in the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.

3.4 Parts, materials and processes: general. Materials,parts and processes shall be in accordance with MIL-P-11268.

3.4.1 Radioactive material restriction in optical products. Optical products shall contain no thorium or other source materials as defined by Title 10, Code of Federal Regulations, Part 40, in excess of 0.05 percent by weight (500 ppm) or any other added radioactive materials.

3.5 Components. The LR Set shall consist of the following components:

1. Laser Infrared Observation Device MX-9838/GVS-5
2. Case, Field, Laser Infrared Observation Device CY-7536/GVS-5

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3. Case, Transit, Laser Infrared Observation Set CY-7535/GVS-5
4. Mount, Adapter MT-4832/GVS-5
5. Cable Assembly, Special Purpose, Electrical CX-13021/GVS-5

3.6 Performance characteristics. The LR shall have the following performance characteristics when operated with the internal battery or Special Purpose Electrical Cable Assembly CX-13021/GVS-5 and external power sources having characteristics as specified in MIL-STD-1275, no fault condition.

3.6.1 Pulse width. The pulse width shall not be less than 4 nor greater than 8 nanoseconds measured at half power points.

3.6.2 Energy output. The energy output shall be ten (10) millijoules minimum within a 1.0 milliradian diameter beam.

3.6.3 Beam divergence. Not less than 80 percent of the output energy shall be contained within a 1 milliradian diameter beam divergence as measured in the far field.

3.6.4 Range. The LR shall determine and display the distance of a target from 200 meters to 9990 meters in 10 meter increments.

3.6.4.1 Target ranging. The LR shall, with 99 percent or greater probability of detection, range to targets normal to the laser beam which subtend an angle not less than 1 milliradian, have a diffused Lambertian reflectance not greater than 10 percent at 1065 nanometers and a visual range of 5 kilometers (atmospheric two-way tranmission 0.01 at 1065 nanometers).

3.6.4.2 Range error. The range error shall be $\pm$ 10 meters RMS maximum.

3.6.5 False alarm rate. The false alarm rate, on the average, shall be less than one false range reading in one hundred (100) valid range measurements, under the maximum range conditions (9980m).

3.6.6 Minimum range setting. The LR minimum range shall be continuously adjustable from a low limit of 170 to 190 meters to a high limit of 5000 to 5300 meters. Minimum range (MIN-RNG) setting shall be displayed in the range readout when the MIN. RNG switch is depressed.

3.6.7 Target detection. The LR shall detect the first target beyond 10 meters from the minimum range setting when tested as specified herein.

3.6.8 Target discrimination. The LR shall discriminate between targets along the line-of-sight where range separation is 100 meters or greater and shall give a visual indication for multiple targets.

3.6.9 Probability of detection. The probability of detection shall not be less than 99 percent.

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3.6.10 Boresight. The boresight error between the transmitter and receiver sighting telescope shall not exceed 0.5 milliradian.

3.6.11 Duty cycle. The LR shall be capable of 12 rangings in a period of 2 minutes (6 pulses per minute) followed by 4 rangings in 8 minutes (1 pulse per 2 minutes) with cycle capable of being repeated indefinitely.

3.6.12 System response. The period between activating the ranging sequence and displaying of the target range shall not exceed one (1) second. The target range shall be displayed until the operator deactivates the ranging sequence.

3.6.13 System recycling. System recycling shall be automatic between ranging sequences.

3.6.14 Battery indicator. A visual indication shall be provided to the operator at the end of the useful charge in the battery. The indicator shall be activated when the battery voltage drops to 21.5 ±1 Vdc.

3.6.15 Low voltage cut-off. An automatic low voltage cut-off shall remove power from all circuits when the power source voltage at the LR is below 17.5 ±1.5 volts.

3.6.16 Flash lamp stray emission. All visual flash lamp light shall be contained within the equipment housing.

3.6.17 Number of ranging operations per battery charge. There shall not be less than 100 rangings (based on five (5) second range display), over the operating temperature range of the BB-516/U battery.

3.7 Environmental conditions.

3.7.1 Altitude. The LR shall not be damaged (see 6.3) during operation at altitudes up to 10,000 feet above sea level.

3.7.1.1 Altitude transportation. The LR Set shall not be damaged by transportation up to 40,000 feet above sea level in unpressurized aircraft while in its transit case.

3.7.2 High temperature. The LR shall not be damaged by storage to +71°C and operation to +71°C.

3.7.3 Low temperature. The LR shall not be damaged by storage to -57°C and operation to -46°C.

3.7.4 Humidity. The LR and Cable Assembly CX-13201/GVS-5 shall not be damaged by exposure to 94 ±4 percent relative humidity at +65°C maximum ambient air temperature, when tested as specified herein.

3.7.5 Fungus. The LR shall be resistant to fungus growth.

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3.7.6 Vibration. The LR Set shall not be damaged by vibrations encountered during transportation as loose or secured cargo. The LR in its field case shall not be damaged by vibrations encountered during transportation as loose cargo.

3.7.7 Shock. The LR Set shall not be damaged by shocks encountered during transportation, bench handling and servicing.

3.7.8 Leakage (Immersion). The LR shall not be damaged when immersed in fresh water to a depth of 36 ±5 inches for a period of 120 ±5 minutes.

3.7.9 Acoustical noise (Aural non-detectability test). The LR shall comply with the requirements of MIL-STD-1474 Para 5.2.

3.8 Electromagnetic interference (EMI). The LR shall comply with the following requirements of MIL-STD-461A, Notice 4, Table VI, for Class Number ID equipment during operation on internal and on external power using the external power cable CX-13021/GVS-5.

Internal Battery Operation

RE02	Radiated Emissions, Electric Field, Broadband, 14 kHz to 1 GHz
RE02.1	Radiated Emissions, Electric Field, Narrowband, 14 kHz to 10 GHz
RS02	Radiated Susceptibility, Magnetic Induction Field, Spike
RS03.1	Radiated Susceptibility, Electric Field, 2 MHz to 10 GHz

External Power Condition

CE01	Conducted Emissions, DC Power Leads, 30 Hz to 20 kHz
CE04	Conducted Emissions, DC Power Leads, 20 kHz to 50 MHz
CS01	Conducted Susceptibility, DC Power Leads, 30 Hz to 50 kHz
CS02	Conducted Susceptibility, Power Leads, 50 kHz to 400 MHz
CS06	Conducted Susceptibility, Power Leads, Spike
RE02	Radiated Emissions, Electric Field, Broadband, 14 kHz to 1 GHz
RE02 (Modified)	Radiated Emissions, Electric Field, Narrowband (Figure 1 modified to increase the limit at 14 kHz to 66 db μV/meter and to connect new limit by a straight line to the existing limit at 300 kHz.)
RS02	Radiated Susceptibility, Magnetic Induction Field, Spike
RS02.1	Radiated Susceptibility, Electric Field, 2 MHz to 10 GHz

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3.9 Burn-in. Each LR shall have accumulated a minimum of 24 hours operation under the test conditions specified herein.

3.10 Reliability. The LR shall have a specified MRBF (mean rangings between failures) of 30,000 rangings, under the test conditions specified herein.

3.11 Compatibility. The LR shall be compatible electrically and mechanically with the Test Set, Laser Infrared Observation Device TS-3620/GVS-5.

3.12 Finish.

3.12.1 General. The equipment shall be finished in accordance with the equipment drawings.

3.13 Nameplates and product marking.

3.13.1 Nameplates. Each nomenclatured item shall be marked with a serialized nameplate conforming to MIL-P-15024/6 Type A or G.

3.13.2 Marking. The equipment shall be permanently and legibly marked for identification in accordance with MIL-M-13231.

3.14 Government loaned property. Unless otherwise specified in the contract or purchase order, the Government will provide the following items for use in the production of this equipment.

NSN	Nomenclature	Quantity
	Test Set, Laser Infrared Observation Device TS-3620/GVS-5	2
	Charger Battery, PP-7286/U	1
	Battery BB 516/U	24

3.15 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.5).
c. Inspection of packaging (see 4.13).

4.3 First article inspection. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 Inspection. Each first article LR Set shall be inspected in accordance with Table I, both in process and when assembled, as required. Presence of one or more defects shall be cause for rejection of that LR Set and may be cause for rejection of first article.

Table I. Inspections.

Inspection Requirement	Requirement paragraph	Inspection criteria
Any part or component missing or damaged	3.2, 3.4	MIL-STD-252
Weight not as specified	3.2.1	MIL-STD-252
Marking not as specified	3.13	MIL-M-13231
Workmanship not as specified	3.15	MIL-STD-454
Treatment and painting not as specified	3.12	Equipment Drawings

4.3.2 Tests. Upon successful completion of the inspections specified in 4.3.1, the first article LR Sets shall be subjected to Burn-in (paragraph 4.9) and to the tests in table II. Prior to start of tests, the LR Sets shall be identified as to their applicable subgroups (see 6.2). Tests may be conducted in any order except as noted in table II. Failure of any test shall constitute first article failure.

TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph	Subgroup 1	2	3
Pulse width	3.6.1	4.6.1	X	X	X
Energy output	3.6.2	4.6.2	X	X	X
Beam divergence	3.6.3	4.6.3	X	X	X
Range	3.6.4	4.6.4	X	X	X
False alarm rate	3.6.5	4.6.5	X	X	X
Minimum range setting	3.6.6	4.6.6	X	X	X

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TABLE II. First article inspection. - Continued

Inspection	Requirement paragraph	Test paragraph	Subgroup 1	2	3
Target detection	3.6.7	4.6.7		X	X	X
Target discrimination	3.6.8	4.6.8		X	X	X
Probability of detection	3.6.9	4.6.9		X	X	X
Boresight	3.6.10	4.6.10		X	X	X
Duty cycle	3.6.11	4.6.11		X	X	X
System response	3.6.12	4.6.12	X	X	X	
System recycling	3.6.13	4.6.13	X	X	X	
Battery indicator	3.6.14	4.6.14	X	X	X	
Low voltage cut-off	3.6.15	4.6.15	X	X	X	
Flash lamp stray emission	3.6.16	4.6.16	X	X	X	
Ranging operations	3.6.17	4.6.17	X	X	X	
Altitude	3.7.1	4.7.1	X			
Altitude transportation	3.7.1.1	4.7.1.1	X			
High temperature	3.7.2	4.7.2	X			
Low temperature	3.7.3	4.7.3	X			
Humidity	3.7.4	4.7.4	X			
Fungus 1/	3.7.5	4.7.5			X	
Vibration 2/	3.7.6	4.7.6	X			
Shock 2/	3.7.7	4.7.7	X			
Leakage (immersion) 2/	3.7.8	4.7.8	X			
Acoustical noise 1/	3.7.9	4.7.9			X	
Electromagnetic interference 1/	3.8	4.8			X	
Reliability	3.10	4.10		X		
Compatibility	3.11	4.11	X			

1/ Unless otherwise specified in the contract or purchase order, or major design changes or changes in components, materials or supplies during subsequent production are made, fungus, acoustical noise and EMI tests apply to first article only.

2/ Vibration and shock shall be conducted before leakage.

4.4 Inspection procedure. Unless otherwise specified herein, the LR shall be operated at an input voltage of 24.5 ±2.5 Vdc at an ambient temperature of 23°C ±5°C. All specified tolerances are absolute values and errors due to test equipment shall be subtracted from those tolerances.

4.5 Quality conformance inspection.

4.5.1 Inspection. Inspection shall be as specified in 4.3.1.

4.5.2 Tests.

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4.5.2.1 Group A inspection. Group A inspection shall be conducted on all LR units which have been subjected to burn-in specified in 4.9. Group A tests listed in table III may be performed in any order. Failure of any test shall be cause for rejection of that unit.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Energy output	3.6.2	4.6.2
Beam divergence	3.6.3	4.6.3
Range	3.6.4	4.6.4
False alarm rate	3.6.5	4.6.5
Target detection	3.6.7	4.6.7
Probability of detection	3.6.9	4.6.9
Boresight	3.6.10	4.6.10
Duty cycle	3.6.11	4.6.11

4.5.2.2 Group B inspection. Group B inspection shall be conducted on LR's selected from units which have passed the tests in 4.5.2.1. Sampling of lots (see 6.3) shall be in accordance with MIL-STD-105, AQL 2.5 percent defective, inspection level S-3 combined. Group B tests listed in table IV may be performed in any order.

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Test paragraph
Pulse width	3.6.1	4.6.1
Minimum range setting	3.6.6	4.6.6
Target discrimination	3.6.8	4.6.8
System response	3.6.12	4.6.12
System recycling	3.6.13	4.6.13
Battery indicator	3.6.14	4.6.14
Low voltage cut-off	3.6.16	4.6.16
Ranging operations	3.6.15	4.6.15
Compatibility	3.11	4.11

4.5.2.3 Group C inspection. Group C inspections shall be conducted on LR Sets selected from units which have passed Group A, or Group A and B inspections. The LR Sets shall be tested in accordance with the inspections listed in table V. Samples shall be selected in accordance with 4.5.2.3.1. Tests may be performed in any order, except as noted in table V.

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4.5.2.3.1 Sampling for group C inspection. The acceptance test shall be performed on all inspection lots (see 6.3) except the first article inspection lot. The sample size shall be 3 units. The sample shall be randomly selected from units produced during the sampling period.

TABLE V. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Altitude	3.7.1	4.7.1
Altitude transportation	3.7.1.1	4.7.1.1
High temperature	3.7.2	4.7.2
Low temperature	3.7.3	4.7.3
Humidity	3.7.5	4.7.5
Vibration 1/	3.7.6	4.7.6
Shock 1/	3.7.7	4.7.7
Leakage 1/	3.7.8	4.7.8

1/ Vibration and shock shall be conducted before leakage.

4.5.2.3.2 Group C failures. Actions required relative to group C failures shall be specified in the contract or purchase order (see 6.2).

4.5.2.3.3 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspections may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5.2.4 Group D inspections. Group D inspection shall be conducted on LR's selected from units which have passed Group A, or Group A and B inspections. Test shall be conducted in accordance with table VI.

TABLE VI. Group D inspection.

Inspection	Requirement paragraph	Test paragraph
Reliability	3.10	4.10

4.5.2.4.1 Sampling for group D inspection. The reliability acceptance test shall be performed on all inspection lots (see 6.3) including the First Article inspection lot. The sample size shall be 2 units. The sample shall be randomly selected from units produced during the sampling period.

4.5.2.4.2 Group D failures. Actions required relative to group D failures shall be as specified in the contract or purchase order (see 6.2)

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4.5.2.4.3 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection. As a minimum, the Flash Lamp SM-D-852512 shall be replaced.

4.6 Test methods. Unless otherwise specified all tests shall be conducted at ambient temperatures (see 4.4). The test methods specified are representative. Test methods or equipment which can be demonstrated to be equivalent may be used.

4.6.1 Pulse width. Position a fast response radiometer or detector sensitive to 1065 nanometer radiation to receive the transmitted pulse from the LR. Connect the output of the detector to a fast response storage oscilloscope or oscilloscope equipped with a Polaroid camera. Cover the receiver aperture, then fire the LR. Determine the pulse width at half amplitude from the stored image. Failure to meet the requirements of 3.6.1 shall constitute failure of this test.

4.6.2 Energy output. On a stable optical bench, position a radiometer, calibrated for pulse energy at 1065 nm, behind a one milliradian diameter aperture plate at the focus of a long focal length lens with sufficient aperture to accommodate the LR sighting/receiving aperture and the transmitter aperture. Center the LR sighting circle on the aperture plate and cover the LR sighting/receiving aperture. Adjust the aperture/LR position for maximum radiometer reading, then fire the LR five times, recording the radiometer readings. Average the five readings and convert the average to output energy using suitable calibration factors. Failure to meet the requirement of 3.6.2 shall constitute failure of this test.

4.6.3 Beam divergence. Using the set-up of 4.6.2, remove the aperture plate so that the effective radiometer aperture is at least 5 milliradians. Cover the LR receiver aperture and fire the LR five times, recording the radiometer readings. Average the five readings. Divide the average of the readings of 4.6.2 above by the average of the readings of this test. Multiply the result by 100 to determine percentage of energy falling within one milliradian. Failure to meet the requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Range. A detector sensitive to 1065 nm radiation, a delay circuit capable of providing delays corresponding to ranges from less than 200 meters to at least 9950 meters in 10-meter increments and a short pulse light emitting device emitting 1065 nm radiation at the required MDS of the LR receiver may be used to verify LR range performance. The detector detects the LR transmitter pulse, drives a delay circuit which pulses the light emitting device. Vary the delay to select 10 ranges which exercise all of the digits of the LR display. At each range fire the LR and record the range reading at least three (3) times for a minmum of thirty rangings. Failure to meet the requirements of 3.6.4 shall constitute failure of this test.

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4.6.5 False alarm rate. Using the test device specified in 4.6.4, with the delay corresponding to 9980 meters, fire the LR 100 times at approximately 10 second intervals, measuring the range displayed each time. The one pulse per 10 second rate can be used at 25°C and below. At higher temperatures the rate must be reduced to one per 30 seconds. Two or more measured ranges deviating by more than ±10 meters from the preset range shall constitute failure of this test (see 3.6.5).

4.6.6 Minimum range setting. With the MIN-RNG switch depressed, vary the MIN-RNG Setting knob from fully counterclockwise to fully clockwise. Failure to meet requirements of 3.6.6 shall constitute failure of this test.

4.6.7 Target detection. Review the results of 4.6.4. Any recorded ranges of 000 shall constitute failure of this test (see 3.6.7).

4.6.8 Target discrimination. The test setup of 4.6.4 may be used. It must be capable of providing both a single optical pulse, and two optical pulses separated by 667 nanoseconds (100 meters). In the two pulse mode, the first pulse shall occur at the delay setting of the delay device, and the second pulse shall occur 667 nanoseconds later. With the test device in the single pulse mode, fire the LR into the test device, record the range delay selected, the range readout LR and the state of the LR multitarget LED (on/off). (The multitarget LED should not be illuminated). Repeat with the test device in the two target mode. (The LR should display the same range with the multitarget LED illuminated). Adjust the MIN-RNG readout on the LR to 20 to 30 meters greater than the range displayed, and fire the LR into the test device. (The LR should display a range 100 ±10 meters greater than the range delay selected and the multitarget LED should not be illuminated). Failure of the LR to give the indicated displays shall constitute failure of this test (see 3.6.8).

4.6.9 Probability of detection. Review the results of 4.6.5 above. More than one failure to display the correct range ±10 meters shall constitute failure of this test (see 3.6.9).

4.6.10 Boresight. Using a test setup similar to 4.6.2, a suitable lens and an aperture fixture with alignment crosshairs on it, align the reticle of the LR receiving optics with the aperture fixture, then cover the receiver aperture. Install a piece of exposed polariod film in the aperture and fire the LR. Measure the distance of the center of the "hot spot" burn on the film from the center of the cross-hairs of the aperture fixture. Compute the angular displacement in milliradians (the ratio of this measurement to the focal length of the lens multiplied by 1000). Failure to meet the requirement of 3.6.10 shall constitute failure of this test.

4.6.11 Duty cycle. Review the data of 4.6.5. Operation at the specified repetition rate demonstrates the duty cycle (3.6.11).

4.6.12 System response. Using a stopwatch and the test device of 4.6.4, depress the fire button of the LR to its second position as you start the stopwatch. Stop the stopwatch when a range is displayed. Record the time required for the LR to fire. Failure to meet requirement of 3.6.12 shall constitute failure of this test.

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4.6.13 System recycling. Using the test device of 4.6.4, depress the fire button of the LR to its first detent. Note and record that first one zero, then 3 zeros are displayed in the LR range display. Depress the fire button to the second detent. The LR should fire. Note and record the range delay selected and the range displayed. Release the fire button. Note and record that the LR range display goes off. Depress the fire button to its second detent. The LR should fire and display a range. Failure to meet the requirements of 3.6.13 shall constitute failure of this test.

4.6.14 Battery indicator. Install the CX-1321/GVS-5 Cable Assembly, modified to facilitate measuring the voltage at the battery contact points in the LR. Attach a laboratory power supply to the cable input. Adjust the power supply to provide an input of 24 ±2.5 volts when measured at the battery contact points in the LR. Turn the LR power switch on and depress the MIN-RNG button. A range should be displayed and the LOW-BATT light should be off. Holding the LR MIN-RNG button depressed, decrease the power supply voltage until the LOW-BATT light is brightly illuminated. Note and record the voltage at the LR. Failure to meet the requirements of 3.6.14 shall constitute failure of this test.

4.6.15 Low voltage cut-off. Using the test set-up of 4.6.14, depress the LR MIN-RNG button and decrease the power supply voltage from 24 Vdc until the range display in the LR goes out. Note and record the voltage at the LR at which this occurs. Failure to meet the requirement of 3.6.15 shall constitute failure of this test.

4.6.16 Flashlamp stray emission. Cap the LR receiving aperture to avoid damage to the detector from reflected laser energy and insert an opaque plug in the LR eyepiece to block radiation from the range display. Aim the LR at a diffuse black surface. Position the observers so that they do not view the output window surface of the LR and the reflecting surface at which it is fired. The area in which the test is conducted shall have an ambient illumination of not more than 10^{-4} lumens/m^2. Three observers having normal vision or vision corrected to 20-20 shall have been dark-adapted for 30 minutes in the test area at the time of the test. The observers shall view the LR when it is fired and note presence or absence of visible light. Failure to meet the requirements of 3.6.16 shall constitute failure of this test. CAUTION: THIS TEST IS HARARDOUS, EXERCISE EXTREME CARE WHEN CONDUCTING IT.

4.6.17 Number of ranging operations per battery. Energize the LR from an external source of 24.0 ±0.5 V power through a precision 0.1 ohm current sampling resistor in the negative lead. Using a storage oscilloscope connected to the sampling resistor, fire the LR by fully depressing the FIRE button. Release the FIRE button five seconds after the LR laser fires. Compute the current x time product from the oscilloscope waveform in milliampere seconds. Multiply by 100, number of shots required, and divide by 3600 to obtain total milliampere hours required by 100 shots. Milliampere hours exceeding fifty (50) constitute failure of this test.

4.7 Environmental tests. Unless otherwise specified: (a) the environmental tests shall be performed in accordance with MIL-STD-810, as modified by this specification, (b) operational tests required before, during

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and after the environmental test conditions shall be those specified in 4.6.2, 4.6.3, 4.6.4, 4.6.5, 4.6.7, 4.6.9, and 4.6.10. (c) an external power source and the CX-13021 Cable Assembly shall be used, (d) not less than 70 percent of the output energy shall be contained within a 1.0 milliradian beam at temperature extremes.

4.7.1 Altitude. The LR shall be tested in accordance with MIL-STD-810, Method 500.1, Procedure I except the chamber pressure of Step 2 shall correspond to an altitude of 10,000 feet above sea level. Tests requireing mounting of the LR on an optical alignment fixture may be performed with the case of the LR overpressurized to simulate an altitude differential of 10,000 feet, in lieu of being conducted in an altitude chamber. Failure to meet the requirements of 3.7.1 shall constitute failure of this test.

4.7.1.1 Altitude transportation. The LR Set shall be tested in its transit case in accordance with MIL-STD-810, Method 500.1, Procedure I, except the chamber pressure of Step 2 shall correspond to an altitude of 40,000 feet above sea level (Step 2 operation not applicable). Failure to meet the requirements of 3.7.1.1 shall constitute failure of this test.

4.7.2 High temperature. The LR shall be tested in accordance with MIL-STD-810, Method 501.1, Procedure II. In Step 8, the tests of paragraph 4.7b shall be performed. Failure to meet requirements of 3.7.2 shall constitute failure of this test.

4.7.3 Low temperature. The LR shall be tested in accordance with MIL-STD-810, Method 501.1, Procedure I, except the time specified in Step 2 shall be two hours. In Step 5, the tests of paragraph 4.7b shall be performed. Failure to meet the requirement of 3.7.3 shall constitute failure of this test.

4.7.4 Humidity test. The LR and Cable Assembly CX-13021/GVS-5 shall be tested in accordance with MIL-STD-810, Method 507.1, Procedure II, except operation during Step 6 shall not be required. Accumulated moisture may be wiped off the optical surfaces. Discoloration, spotting, or other visible alterations in the appearance of the externally coated optical surfaces, not affecting final performance (Steps 8 and 9) shall not be cause for rejection. Failure to otherwise meet the requirements of 3.7.5 shall constitute failure of this test.

4.7.5 Fungus. The LR shall be tested in accordance with MIL-STD-810, Method 508.1, Procedure I. The optical surfaces shall not be sprayed. Operation of the equipment is not required. At the conclusion of the exposure period, a visual examination shall reveal no evidence of visible fungus or corrosion. Failure to otherwise meet the requirement of 3.7.5 shall constitute failure of this test.

4.7.6 Vibration. The LR Set shall be tested in accordance with MIL-STD-810 as follows: (a) The LR Set, stowed in the transit case, shall be subjected to Method 514.2, Procedures X and XI, Part 2 of MIL-STD-810. (b) The LR, in its field case shall be subjected to Method 514.2, Procedure XI, Part 2 of MIL-STD-810. The vibration level shall be in accordance with Figure 514.2-7, curve AW,

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and the sweep time shall be in accordance with Table 514.2 - VII of MIL-STD-810. Failure of any test to meet the requirement of 3.7.6 shall constitute failure of that test. Dents and finish damage to the case or case inserts shall not be cause for rejection. Cases with such damage may be reused for tests.

4.7.7 Shock. The LR Set shall be tested in accordance with MIL-STD-810 as follows: (a) The LR Set stowed in its transit case shall be subjected to Method 516.2, Procedure II of MIL-STD-810. (b) The LR shall be subjected to Method 516.2, Procedure V of MIL-STD-810. Failure to meet the requirement of any test shall constitute failure of that test. Dents and finish damage to the case or case inserts shall not be cause for rejection. Cases with such damage may be reused for tests.

4.7.8 Leakage (immersion). The LR and the transit case shall be separately tested in accordance with MIL-STD-810, Method 512.1, Procedure I. Pre-test and post-test operation (see 4.7) shall be performed. Prior to the post immersion operational test, the external optical surfaces may be wiped dry. Failure to meet the requirements of 3.7.8 shall constitute failure of this test.

4.7.9 Acoustical noise. The octave-band sound pressure measurements shall be made in an environment having free field characteristics and noise levels being measured. The LR shall be installed and operated in a vertical position. A survey shall be first made to determine the location, on a circle of 1.25 meter radius from the center of the equipment, of maximum sound output. The octave band analysis shall be made at this point on the circle. Measurements shall be made with equipment in MIN RNG and FIRE modes using an octave band noise analyzer whose characteristics comply with USASI Publication S1.4 and S1.11 (see 3.10.1.10).

4.8 Electromagnetic interference. The LR shall be tested in accordance with MIL-STD-462, Notice 3 (EL) for compliance with the conducted and radiated emission and susceptibility requirements specified in 3.8. The LR shall be tested during steady state operation (Power switch at PWR ON, MIN RNG and RET ON switches depressed) with MIN RNG SET and RET BRT controls set to mid position and also with the MIN RNG SET and RET BRT controls set to their maximum. Susceptibility only shall be measured at the latter minimum range setting. The LR shall also be tested during introduction of transient conditions by alternately depressing and releasing the push button switches. Failure to meet requirements of 3.8 shall constitute failure of this test.

4.9 Burn-in. Each LR shall be subjected to the burn-in cycle specified in Figure 2. Input voltage shall be 24.5 $\pm$2.5 volts dc. Vibration will be 2.2G + 10 percent peak acceleration at any non-resonant frequency between 20 and 60 Hz measured at the mounting points on the equipment. The duration of vibration shall be 30 minutes with the equipment not operating. Failure of burn-in shall be defined as failure to pass the inspections listed in table III. Any unit which fails and is repaired or is later altered, repaired or reworked, shall be subjected to the specified burn-in cycle.

4.10 Reliability. Unless otherwise specified in the contract the reliability test shall be run.

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4.10.1 Reliability test. The sample of 2 units shall be tested for 54,000 ranges with neither unit experiencing fewer than 18,000 ranges. Successful completion of the test occurs when there are three or fewer failures. The units shall be monitored at least every 24 hour period, using the operational tests specified in 4.6.2, 4.6.3, 4.6.4, 4.6.5, 4.6.7, 4.6.9 and 4.6.10 to ensure the units conform to this specification. Any nonconformance shall be classified a failure. Ranges from the period in which the failure has occurred shall not count toward the 54,000 ranges. A failed unit shall be removed from test for failure analysis. The unit shall be repaired and returned to test.

4.10.2 Reliability test conditions. The units shall operate under the following conditions:

Temperature: 50° C + 5° - 0°(122°F to 131°F).

Vibration 2.2G $\pm$ 10 percent peak acceleration.

Vibration at any non-resonant frequency between 20 and 60 Hz measured at the mounting points on the equipment. The duration of the vibration shall be 30 minutes for every 24 hours of testing without operation of the equipment.

Equipment on-off cycling - the equipment shall be cycled continuously at a cycling rate of 6 rangings per minute for 2 minutes followed by one ranging every 2 minutes for the next 8 minutes. Input voltage shall be 24.5 $\pm$ 2.5 volts.

4.10.3 Failure criteria. A failure shall be defined as any occurrence which causes the equipment performance to deviate from the limits specified in this specification and which cannot be corrected by controls accessible during normal use. All failures occurring during reliability testing shall be classified as either relevant or non-relevant.

a. Relevant failures include:

(1) Design/Workmanship failures.
(2) Component part failures which affect system performance.
(3) Intermittent failures.
(4) Adjustments using controls inaccessible during normal use.
(5) Moisture inside system.

b. Nonrelevant failures include:

(1) Failures attributable to improper installation in the test chamber.
(2) Failures attributable to test instrumentation or monitoring equipment.
(3) Failures resulting from test operator error in setting up, or in testing the equipment.
(4) Failures attributable to an error in the test procedure.
(5) Failures clearly attributable to an overstress condition caused by a condition external to the system under test.
(6) Secondary failures as defined in MIL-STD-781 (see 6.2).

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NOTE: At no time after the units are put on test shall the energy output fall to less than 6 millijoules contained in a 1 milliradian beam diameter. A corresponding degradation in range performance not greater than 3 db shall also be permitted.

4.10.4 Failure confirmation. All failures observed during the equipment reliability test shall be confirmed. Failure to do so is insufficient grounds for deletion of such failures in the count of total failures. Lack of failure confirmation should instigate close review of the test facility. If the test facility can be shown to be both at fault and completely accountable for the failure to the satisfaction of the procuring activity, then and only then can the count of such failure be eliminated.

4.10.5 Verifying repair. Following repair or corrective action and prior to the resumption of test, it shall be permissible to operate equipment in the test facility. Test procedures shall specify the period of operation (or number of cycles) which will be used to verify the effectiveness of the repair. Post-repair "burn-in" time will in no case exceed that employed for new equipment acceptance. Failures during this period shall be recorded and reported. Time accumulated during this period shall not be used in determining compliance with requirements. All failures shall be subject to analysis.

4.10.6 Equipment layout. The test system shall be mounted in an environmental chamber.

4.11 Compatibility. The compatibility of the LR with the Test Set TS-3620 shall be demonstrated by mounting the LR on a Test Set. Adjust the MIN RNG knob on the LR to its minimum setting. Set the mode switch on the test set to the 1 TGT position. Press the RESET button on the test set and fire the LR. Verify that the range readout in the LR is between 4600 and 5000 meters and that the MULTI-TGT light is not illuminated. Set the mode switch on the test set to the 2 TGT position and press the RESET button. Fire the LR and verify that the range readout is between 4600 and 5000 meters and that the MILTI-TGT light is illuminated. Adjust the MIN-RNG knob on the LR to read 20 meters greater than the range recorded above. Press the RESET button on the test set and fire the LR. Verify that the range display is 100 $\pm$ 10 meters greater than the range recorded above and that the MULTI-TGT light is not illuminated. Failure to provide the above indications constitutes failure of this test.

4.12 Lens test, thorium content. A test shall be performed to determine that the thorium content of any lens covered in Title 10, Code of Federal Regulations, Part 40.13(c) is below 500 parts/million by weight (see MIL-G-174A). Test procedure shall be as specified in the contract. Results of this test shall be included in the Ionizing Radiation Test Report if required in the contract.

4.13 Quality conformance inspection for packaging. Packaging shall be tested in accordance with MIL-P-116 to determine conformance to the requirements of section 5. The rough handling test will be performed only when specified (see 6.2.f).

5. PACKAGING

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5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. JR-1-1-ØØ-00-NS-X-ED-Ø-00-A.

5.1.2 Level B. JR-1-1-ØØ-00-NS-X-ED-Ø-00-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. Laser Infrared Observation Device AN/GVS-5 is intended to provide first target range information to combat troops. Provisions are made for use with vehicular power sources and for providing bearing and elevation information when used with the proper auxiliary equipment.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When first article tests are to be conducted, it is recommended that three LR Sets be used for subgroup 1, two for subgroup 2 and one for subgroup 3. The subgroup testing may be performed concurrently.
c. Time frame required for submission of first article test reports.
d. Time frame required for approval of first article test reports by the contracting officer.
e. Production delivery schedule defined in terms of lots for group B and C sampling (see 4.5.2.3.1 and 4.5.2.4.1).
f. Level A, level B or commercial preservation and packing (see section 5).
g. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.
h. Actions required relative to group C and group D failures (see 4.5.2.3.2 and 4.5.2.4.2).
i. The contracting officer should arrange to furnish the property listed in 3.14.
j. MIL-STD-781B shall be used in lieu of later editions.
k. Test procedure and acceptable test facilities for Thorium Lens testing specified in 4.12.
l. MIL-STD-461A, Notice 4(EL) shall be used in lieu of later revisions.
m. MIL-STD-462, Notice 3(EL) shall be used in lieu of later revisions.

6.3 DEFINITIONS

6.3.1 Damage. Breakage, loosening, shifting, evidence of corrosion or failure of any finish, hardware, connection or component; and any deterioration in quality or failure to operate within the specification requirements.

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6.3.2 Inspection lot size. An inspection lot is defined as two months contiguous production or 300 units, whichever comes first, of those assemblies which have successfully passed Group A inspection.

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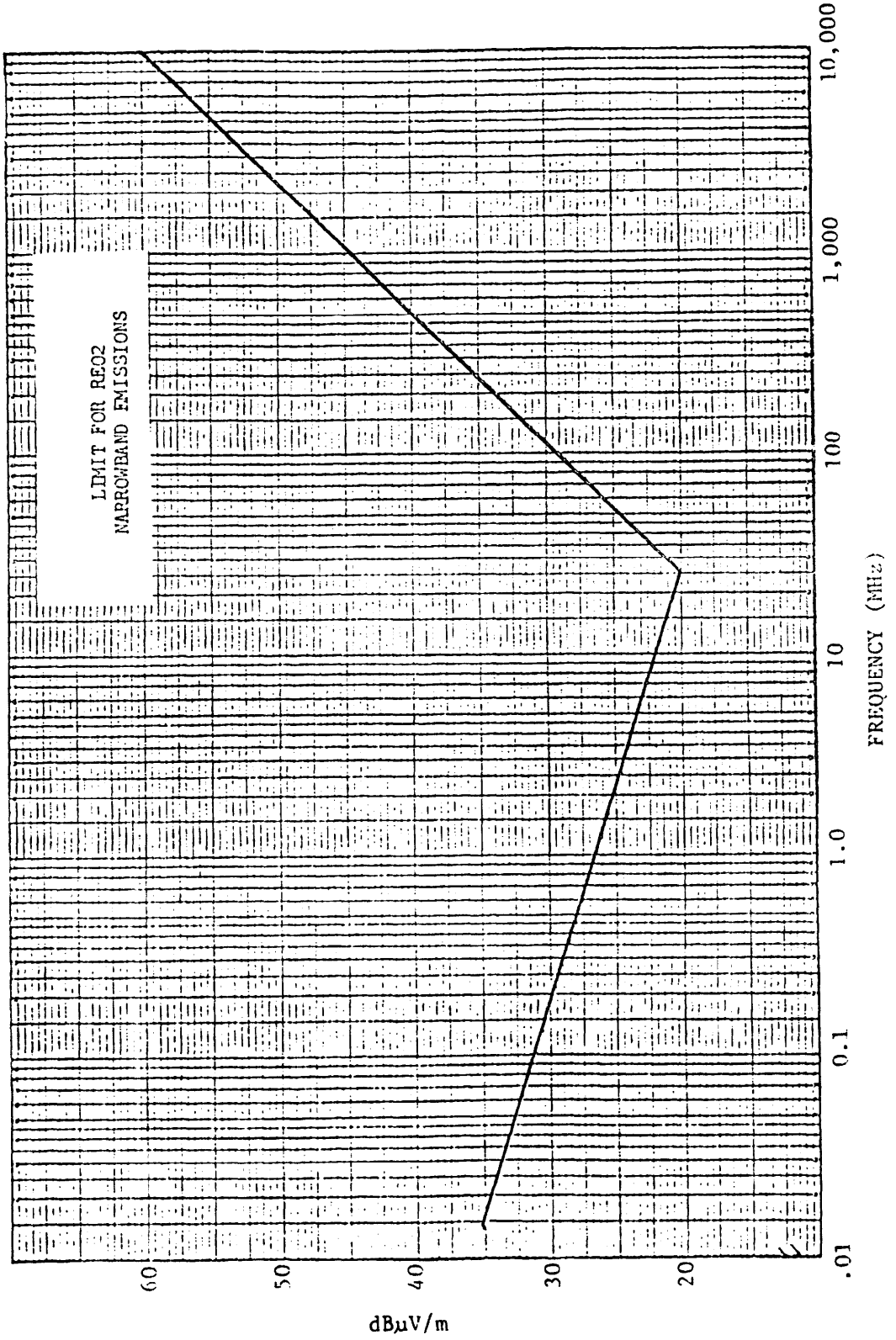


Figure 1. LIMIT FOR RE02 NARROWBAND EMISSIONS

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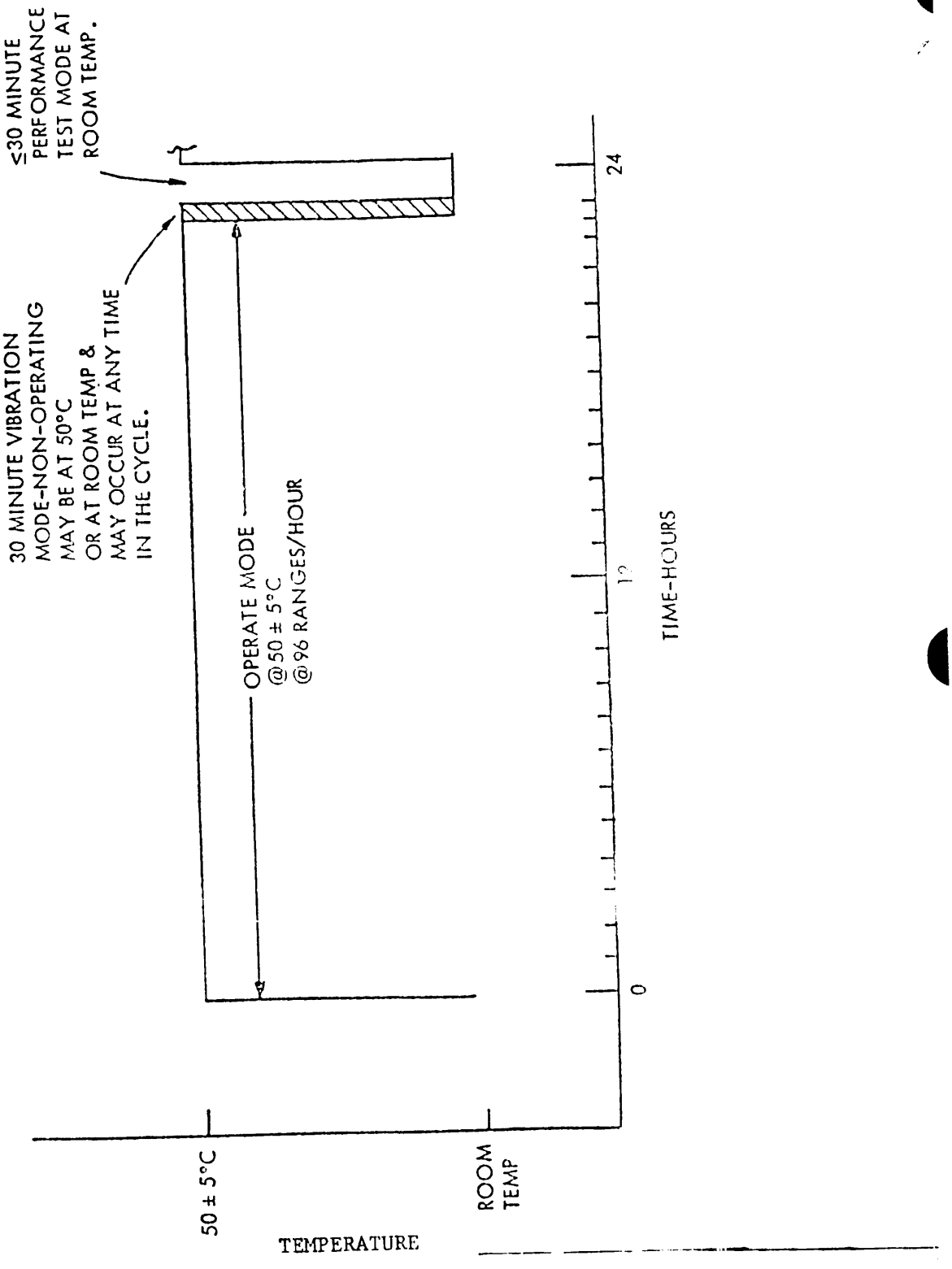


Figure 2. SUMMARY OF TEST CONDITIONS AND CYCLE

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